Proposed Solution

Date	20 July 2026
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine

Project Proposal (Proposed Solution) Report

OptiCrop transforms crop selection using machine learning, boosting agricultural efficiency and productivity. It tackles poor crop selection, inefficient resource use, and low yield caused by environmental uncertainty, promising better farming decisions and improved profitability. Key features include a multi-algorithm crop recommendation model and a simple, real-time web interface for farmers.

Project Overview

Field	Details
Objective	To help farmers, researchers, and agribusinesses identify the most suitable crop for given soil and climate conditions using machine learning.
Scope	The project covers data collection, preprocessing, model training/evaluation (KNN, Logistic Regression, K-Means), and deployment of a web application that delivers real-time crop recommendations.

Problem Statement

Field	Details
Description	Farmers struggle to determine the most suitable crop based on soil nutrients (N, P, K) and climate conditions (temperature, humidity, rainfall, pH), leading to financial losses and reduced yield.
Impact	Solving this improves agricultural productivity, reduces resource wastage, supports sustainable farming, and increases farmer profitability.

Proposed Solution

Field	Details
Approach	Train and evaluate multiple ML algorithms on a labeled crop dataset (soil + climate → crop), select the best-performing model, and serve it through a Flask web application.
Key Features	- Multi-algorithm crop classification (KNN, Logistic Regression) - K-Means clustering for exploratory soil/climate grouping - Real-time prediction via a responsive web interface - Model evaluation and persistence of the best model

Resource Requirements

Resource Type	Description	Specification / Allocation
Computing Resources	CPU specification	Standard laptop CPU (no GPU required for classical ML models)
Memory	RAM specification	8 GB
Storage	Disk space for data, models, logs	~1 GB
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, VS Code
Data	Source, size, format	Crop recommendation dataset (Kaggle), CSV, ~2200 rows × 8

Data	Source, size, format	
		columns (N, P, K, temperature, humidity, pH, rainfall, label)